MURALIDHAR REDDY ANUMANDLA

(804) 616-1946 | anumandlamuralidharreddy@gmail.com | LinkedIn | GitHub | Portfolio
Data Engineer | Richmond, VA | Open to relocation

SUMMARY

Cut a 140M-row Spark job from a 20-hour crash to 2-4 hours. Data Engineer with nearly 1.5 years on Azure Databricks and Data Factory, working on regulated healthcare data at UnitedHealth Group with Kafka streaming, Delta Lake and 25+ monitored pipelines. M.S. Computer Science, VCU.

EXPERIENCE

UNITEDHEALTH GROUP (Client), Remote — Jun 2025 – Present

Data Engineer, Feb 2026–Present | Data Engineer Intern, Jun–Dec 2025

- Fixed a Spark job on 140M rows that crashed out of memory after 20 hours; window-function dedup, early filtering, and Parquet staging now complete it in 2-4 hours.
- Monitor 25+ production ETL/ELT pipelines in ADF and Databricks, running daily, every 4 hours and weekly: post a daily run-status report to a multi-team chat, rerun transient failures, escalate the rest to the team lead.
- Traced duplicate member-coverage records in a continuously running 8-table Kafka pipeline to a faulty derived column and merge key, fixed both, and flagged the source-side cause to the analytics head.
- Built a Kafka-fed Spark Structured Streaming pipeline into Delta tables, decrypting PII via Key Vault and publishing unsubscribe flags that control who the email platform can contact.
- Wrote the one-time query that loaded 3.6M legacy consent records, with the business-required columns, into the new category-level table, fixing a compliance gap that risked wrongly emailing or suppressing users.
- Diagnosed a stalled staging file-ingestion pipeline as a cluster upgrade that broke the pipeline code, fixed the defects it exposed over about 2 weeks, and explained the root cause to business users.
- With a team of four, wrote reusable SQL validation checks, parameterized by table name, and ran them across 20 demographics tables, 6 of them mine, and 8 tables post-migration to verify a deduplication feature.

PERSONAL PROJECTS

NYC Mobility Incremental Pipeline

PySpark, Delta Lake MERGE, dbt, DuckDB, Airflow, GitHub Actions; 48.7M real NYC TLC trips

- Cut a new month's load from 15.6 minutes to 78 seconds on 48.7M NYC taxi trips, by merging only the new data into Delta tables instead of rebuilding everything.
- Got 0 duplicates and 0 differing rows after replaying 19,380 inserted and 4,938 updated late rows, and quarantined 978,679 bad rows, with 37 of 37 dbt checks and CI passing.

CMS Claims Lakehouse

PySpark, Delta Lake, dbt, DuckDB, GitHub Actions; real CMS Medicare public data

- Found a 2.1x state payment gap for hip and knee replacement DRG 470 across 1.56M rows of real CMS Medicare data, by building a bronze, silver and gold lakehouse with dbt star schemas.
- Made re-runs safe, with unchanged counts and 0 duplicate keys, by writing silver with Delta MERGE on the business key; 32 dbt data tests and CI repeat the check.

Sales Analytics Dashboard

Power BI, DAX, DuckDB; 932K e-commerce transactions

- Showed +18.0% and +13.8% yearly revenue growth, by modeling a star schema with 6 DAX measures.
- Found Electronics at 30.1% of volume and 32.1% of profit across 240 products and 40K customers.

SKILLS

Languages:	SQL, T-SQL, window functions, Python, PySpark, Spark SQL
Data Engineering:	Medallion architecture, incremental MERGE loads, Delta Lake, data quality, dbt, Airflow
Azure & Streaming:	Databricks, Data Factory, Unity Catalog, Kafka, Spark Structured Streaming
Engineering Practices:	Git/GitHub, PR workflow, GitHub Actions CI, Agile/Rally, GitHub Copilot, Claude Code
AI & Reporting:	LLM tool-calling, Ollama, vector search, Power BI, DAX, Excel pivots and charts

EDUCATION

VIRGINIA COMMONWEALTH UNIVERSITY, Richmond, VA: M.S. Computer Science, GPA 3.90/4.0 — 2024 – 2025

GRIET, Hyderabad, India: B.Tech Computer Science & Business Systems, GPA 8.31/10 — 2020 – 2024